

Alif Azam

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Research Interest

Computational Fourier Optics · Physics-Informed Neural Networks
Scientific Machine Learning · High-Performance Computing

Skills

Languages	Python, C, Bash Scripting
Packages and Libraries	NumPy, Pandas, Matplotlib,
HPC Tools	Slurm, OpenMPI, OpenMP, CUDA, Git
Operating Systems	Linux, Windows

Education

B.Sc. Hons. in Physics

Shahjalal University of Science and Technology, Sylhet

January 2023-Present

Research Experience

Nonlinear Optics Laboratory

Department of Physics, SUST

December 2024 – Present

- Building a physics-informed spectral enhancement pipeline for label-free cell segmentation in phase-contrast microscopy, combining Fourier-domain quality assessment with PINN-based image restoration and adaptive bandpass filtering.

Projects

Full Diskless Linux HPC Cluster

2025

- Built and deployed a 7-node diskless Linux cluster using PXE boot, DHCP, TFTP, NFS, MUNGE, and Slurm; configured the master node to provide network booting, shared root filesystem, authentication, and workload management for compute nodes.

- Implemented matrix multiplication with OpenMPI and OpenMP to test parallel execution and validate the cluster's distributed and shared-memory computing capabilities.

Nuclear Power Plant Data Analysis

2025

- Analyzed ERA5 temperature data in NetCDF4 format using Python, working with a spatial subset covering Bangladesh and performing data extraction, preprocessing, visualization, and statistical analysis.

Publications

Conference Proceedings

1. Nripendro Biswas, Ahmed Rijvee, Munaem Ahmed Mahdi, **Alif Azam**, Md. Takoat Hossain, “*BSER – Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models.*” Proceedings of the 8th International Conference on Engineering Research, Innovation and Education (ICERIE 2025), Atlantis Press. DOI: 10.2991/978-94-6463-884-4_77

Presentations

1. “*BSER - Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models*”: Nripendro Biswas, Ahmed Rijvee, Munaem Ahmed Mahdi, **Alif Azam** and Md. Takoat Hossain. *International Conference on Engineering Research, Innovation and Education (ICERIE 2025)*.

Linguistic Skill

English (Fluent) · Bangla (Native)

Workshops/Courses

University of Colorado Boulder	link
Introduction to High-Performance and Parallel Computing	June 2026
Google	link
System Administration and IT Infrastructure Services	Jan 2025
Google Digital Marketing Course	July 2021
SoloLearn	
Python Core Course	Feb 2020